

FitLife App: Funnel, Economics & Paywall Optimization

An end-to-end data study analyzing acquisition channels, user cohort retention, unit economics, and A/B test results for subscription growth.

PRODUCT MODEL

Freemium + \$9.99/mo

ANNUAL TRAFFIC

1,200,000 Users

TECH STACK

PostgreSQL • Python • Statsmodels

🎯 Business Context & Objectives

⚠️ Business Challenges

- **Rising Acquisition Costs (CAC):** Ad spend is increasing across paid channels without proportional subscriber gains.
- **Paywall Conversion Bottleneck:** Low conversion rate from 7-day free trial to paid subscription.
- **Early Subscription Churn:** High drop-off in user retention during Months 2 and 3 of life cycle.

📋 Project Objectives

- **Funnel Extraction (SQL):** Identify bottlenecks across acquisition sources (Organic, Google Ads, Instagram, TikTok).
- **Unit Economics (Python):** Build LTV/CAC model to evaluate baseline business viability.
- **Paywall A/B Experiment:** Test redesigned Paywall screen and validate statistical significance (Z-test).
- **Quantify Financial Impact:** Translate statistical uplift into projected annual revenue gain.

Funnel & Channel Efficiency (SQL)

 **Organic Traffic Dominance**

Leads volume with 700 users and highest trial activation rate (13.43%) across all channels.

 **Instagram High Quality**

Generates the highest Trial-to-Paid conversion rate (15.91%), proving strong audience intent.

 **TikTok Critical Issue**

300 acquired users resulted in 0% paid subscribers, signaling invalid traffic or poor targeting.

 **Blended Baseline CR1**

Aggregated overall conversion from visit to paid subscriber stands at a baseline of 1.53%.

	 A-Z source ▼	123 total_users ▼	123 trials ▼	123 paid ▼	123 cr_to_trial ▼	123 cr_trial_to_paid ▼
1	Organic	700	94	12	13.43	12.77
2	Google Ads	500	32	4	6.4	12.5
3	Instagram	500	44	7	8.8	15.91
4	TikTok	300	5	0	1.67	0

Weekly User Cohort Retention

Steep Onboarding Drop

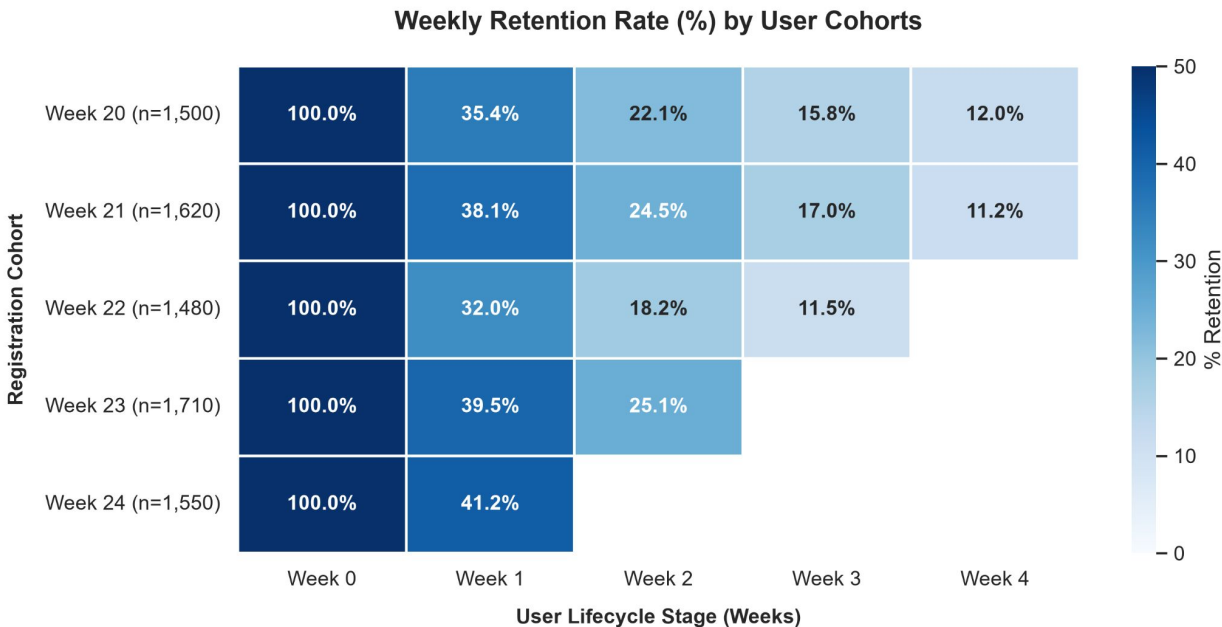
Week 1 retention plunges to 32%–41% across cohorts (~60% user churn in the first 7 days of installation).

Long-term Stabilization

Retention flattens to 11%–12% by Week 4, establishing the core active baseline for surviving subscribers.

Cohort Week 22 Anomaly

Suffered a sharp drop at Week 2 (18.2% vs ~24% avg), indicating possible app technical issues or low traffic quality.



Baseline Unit Economics

Economic Health Assessment: LTV / CAC Ratio = 1.40x (Unhealthy)

The baseline model generates insufficient margin to sustain aggressive user acquisition scales.

\$9.99

MONTHLY AOV

Subscription Price

4.2 mos

AVG LIFESPAN

Active User Duration

\$41.96

CUSTOMER LTV

AOV × Lifespan × 100% Margin

\$30.00

BASELINE CAC

Ad Budget / Buyers

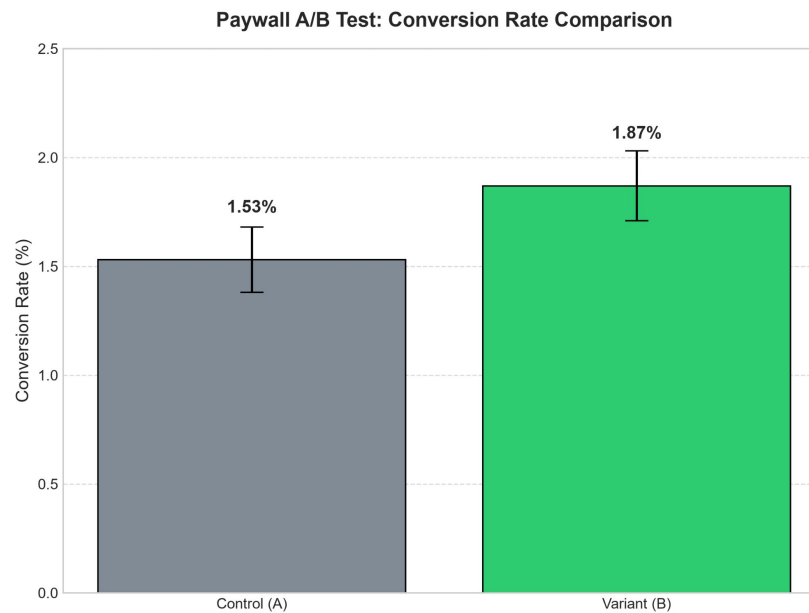
Key Growth Levers:

- 1. Improve CR1 (Trial-to-Paid):** Redesigning Paywall screen to boost initial conversion.
- 2. Extend Lifespan:** Enhance early onboarding to lift retention beyond 4.2 months.

Paywall A/B Experiment Results

+24.7%

RELATIVE CONVERSION GAIN



Statistical Validation

Control (A) CR1: 1.53% (362 buyers / 24,100 users)

Variant (B) CR1: 1.87% (455 buyers / 24,350 users)

Decision & Hypothesis Validation

Z-Test Significance: $Z = 3.303$, $p\text{-value} = 0.00096$ (< 0.05)

Decision: Statistically significant winner! Recommend 100% rollout.

Annual Financial & Business Impact

1.2M

ANNUAL USERS

Projected App Traffic

+4,440

INCREMENTAL BUYERS

22,440 vs 18,000 Baseline

+\$186,258

ADDITIONAL ARR

4,440 buyers × \$41.96 LTV

Improved Unit Economics

Higher conversion rate lowers effective CAC from \$30.00 to \$24.06, boosting LTV/CAC ratio from 1.40x to 1.74x and improving overall ROMI.

Reinvestment Capability

The additional ~\$186k ARR provides margin safety to scale marketing spend while optimizing secondary onboarding funnel stages.

Strategic Growth Recommendations

1. Full Paywall Rollout

Deploy Variant B Paywall screen to 100% of incoming users immediately to capture the +24.7% conversion uplift.

2. Optimize Ad Channels

Reallocate marketing budget away from TikTok (0% paid CR) into Instagram and Organic SEO to maximize high-intent leads.

3. Fix Week 1 Retention

Address the 60% early churn by introducing push notification challenges, personalized workout onboarding, and trial reminders.

Thank You!

FitLife Analytics Case Study — SQL, Python & A/B
Testing

 PostgreSQL SQL Queries

 Python Statsmodels & Seaborn

 Product Analytics